Tracking free-moving bees in an arena

A step-by-step guide to recording bees in an arena, live or from video, and turning their tracks into distances, speeds, time in zones and contact with other bees.

You will learn to

- Set up a camera and arena that track well
- Draw the arena, zones and an edge band
- Set up detection and check it found every bee
- Run live trials, with or without an LED protocol
- Track a saved video
- Read the results and charts, and export your data

You will need

- A laptop or lab PC with Chrome or Edge
- Bee Tracker: the desktop app (BeeTracker.exe) on the lab PC, or `bee-tracker.html`
- A camera on a stand above the arena, or a saved video (MP4 works best)
- The arena's inner width in millimetres
- About 30 minutes for a first run, after watching the 3-minute walkthrough

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Screenshots show a simulated camera with three bees, a food zone and a short LED protocol, recorded by `tools/walkthrough.py`.

PART 1

What Bee Tracker measures

Bee Tracker finds each bee's centre in every sample, follows each bee from sample to sample, and turns the tracks into measures of activity, place preference and contact between bees. It runs in the browser and uploads nothing.

COLUMN	WHAT IT MEANS
Found	The share of samples in which the bee was detected. Below 80% the row is flagged: check detection.
Tracked s	Seconds of continuous track the other measures are based on. Gaps longer than 0.5 s are not counted.
Dist mm	Distance walked, from the smoothed track.
Speed mm/s, 95th	Mean walking speed over the tracked time, and the 95th percentile (a bee's fast walking speed).
Moving	The share of tracked time the bee moved faster than 3 mm/s (set under Results → Settings).
Zone columns	The share of tracked time inside each zone, with the number of visits in brackets.
Edge	Time inside the edge band you set: near the wall (thigmotaxis).
Nearest mm	Mean distance to the nearest other bee. Shown only with two or more bees.
Near bee	The share of time the bee was within 10 mm (centre to centre, about one bee length) of another bee. Shown only with two or more bees.
Encounters, Mean enc. s	How many separate times a bee was near another bee, and how long each lasted on average.
Crossings, Transit s	With linked arenas (zones marked *Arena area*): how often the bee moved from one arena to another, and the time spent between them.
Merges, Reacq.	How often bees touched and showed as one blob, and how often a lost bee was picked up again. Both are warnings that per-bee identity may have swapped.

Per bee or all bees? When unmarked bees touch, Bee Tracker can't always tell which is which afterwards. The **All bees** row doesn't depend on identity and is always safe to use. Use per-bee rows only when each bee is alone in its arena, or the bees are paint-marked and never merged.

PART 2

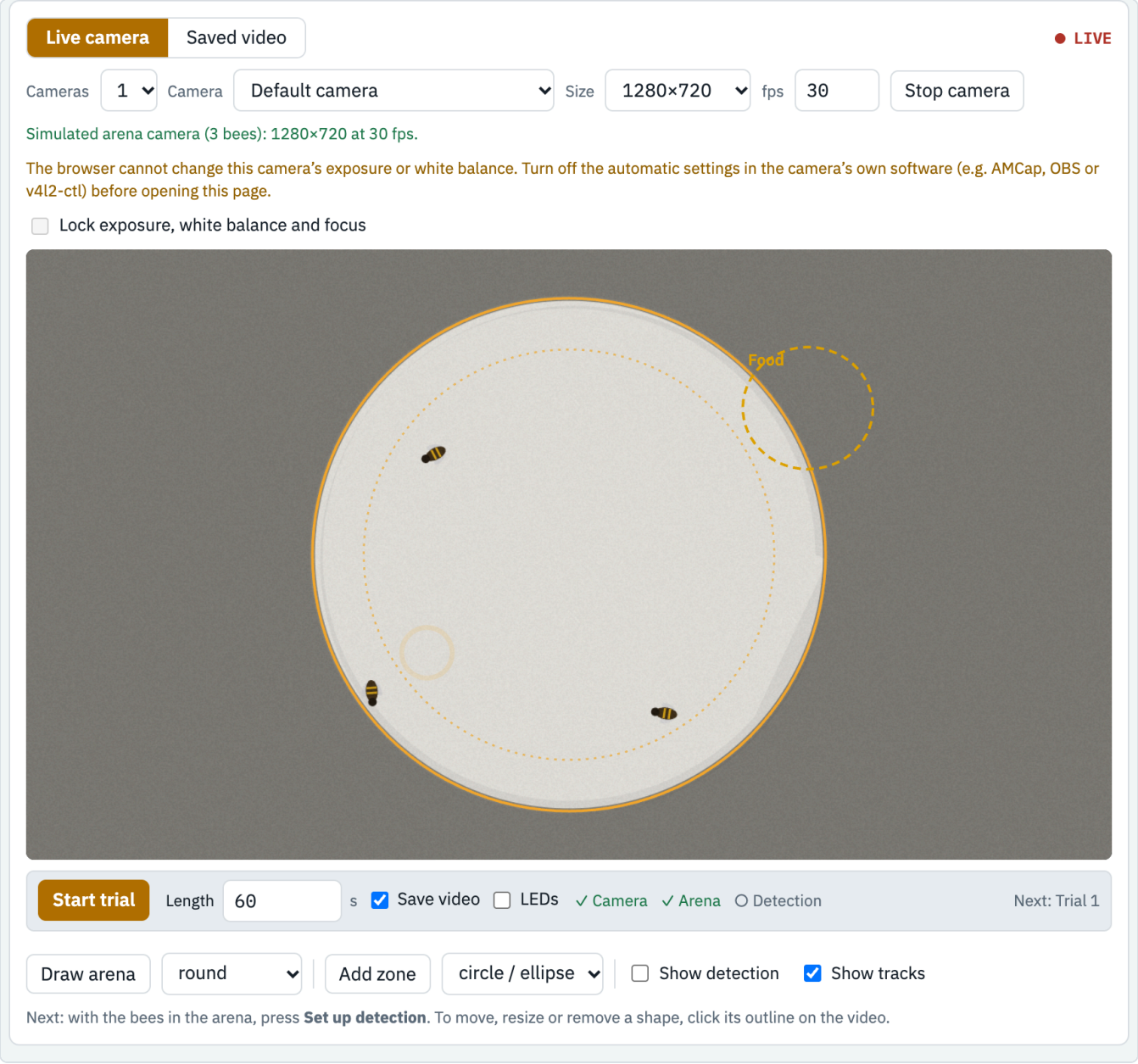
Setting up the camera and arena

The setup matters more than any setting. A clean recording tracks well with the defaults; a poor one can't be fixed afterwards.

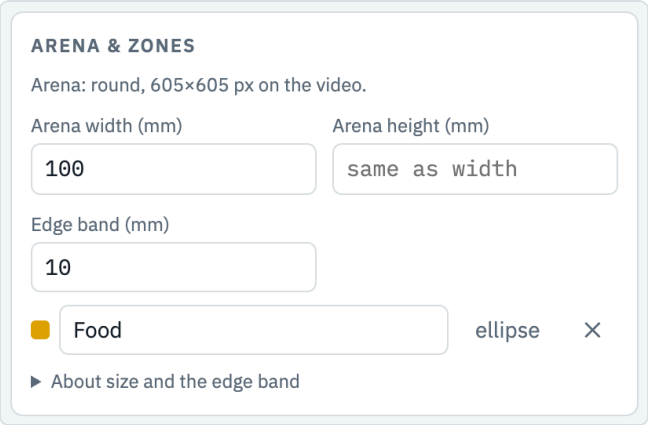
- **Fixed camera, looking straight down,** on a stand, not handheld. If the camera's own software allows it, turn off auto-exposure, auto white balance and autofocus.
- **Even, diffuse light** with no hard shadows. A shadow is a dark blob that moves with the bee.
- **A plain background that contrasts with the bees,** such as white paper or white acrylic under dark bees. Avoid glare off the lid.
- **Lid on the arena.** Bees on the walls or lid, or flying, give bad tracks.
- **Measure the arena's inner width** in millimetres. It sets the scale for every distance.
- **Keep settings the same** across anything you compare: camera size, samples per second and smoothing all change the distance measured.

PART 3

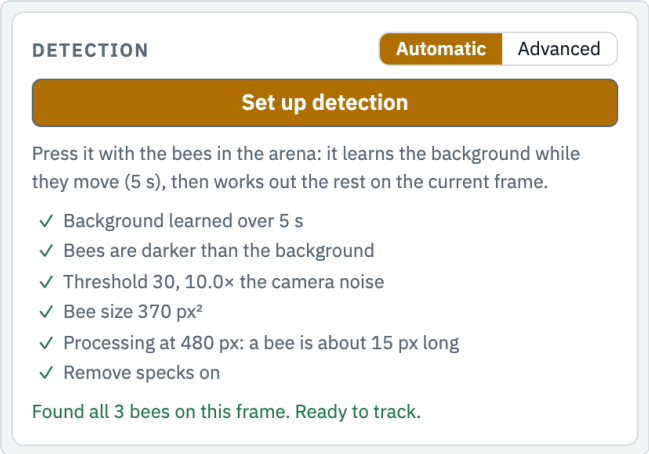
A tour of the screen



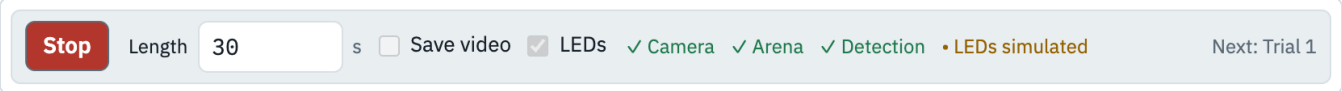
Video. Live camera or saved video. The orange outline is the arena, the dashed circle a zone, the dotted line the edge band. Under the video: the trial bar, then *Draw arena*, *Add zone*, *Show detection* and *Show tracks*.



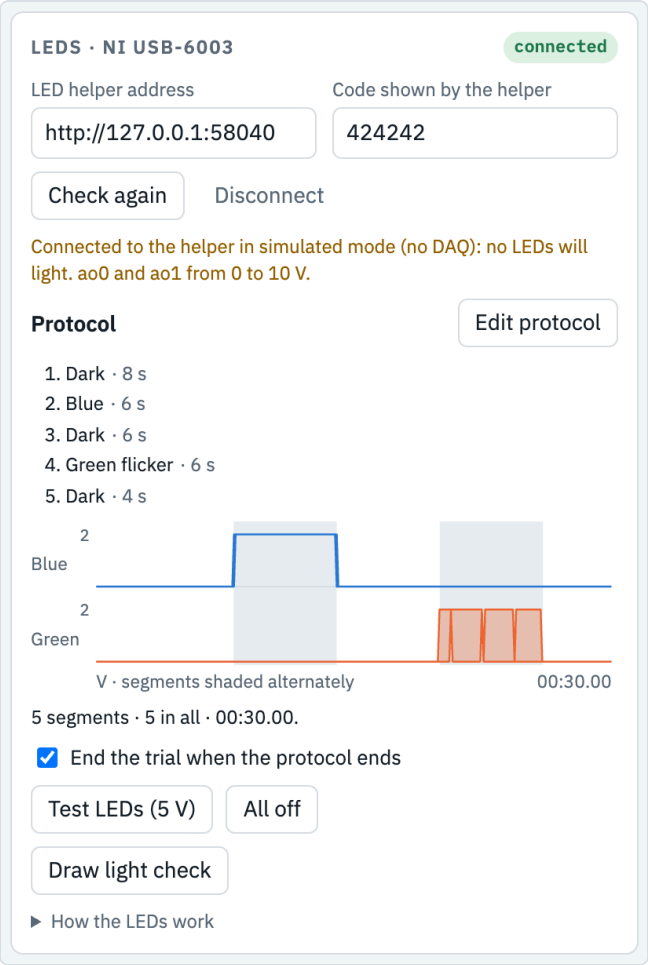
Setup tab. The recording and its group, the number of bees, the arena's real size, the edge band, and the zones.



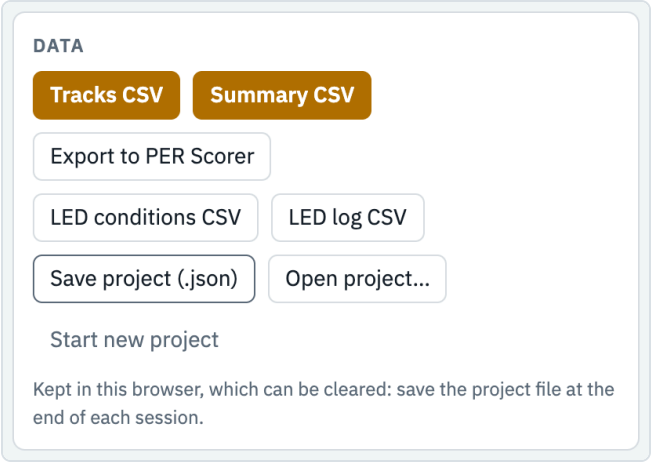
Detect tab. *Set up detection* and what it chose. *Advanced* has every detection and tracking setting.



Trial bar. Length, *Save video*, *LEDs*, and what's ready. During a trial, *Start trial* becomes a red *Stop*. For a saved video the bar has *Start here*, *End here* and *Track bees* instead.



LEDs tab (live camera only). The connection to the LED helper, the protocol, its preview, and *Test LEDs (5 V)*.



Data tab. CSV exports, the PER Scorer export, and saving or opening the project file. The `?` at the top right lists the keyboard shortcuts.

PART 4

Running a live trial, step by step

1 Open Bee Tracker and fill in the session

On the lab PC, start **BeeTracker.exe**: it opens the page in Chrome or Edge, already connected to the LEDs. Keep its small window open while you work. Elsewhere, open `bee-tracker.html`. At the top, type your initials, check the date, and name the experiment.

2 Start the camera

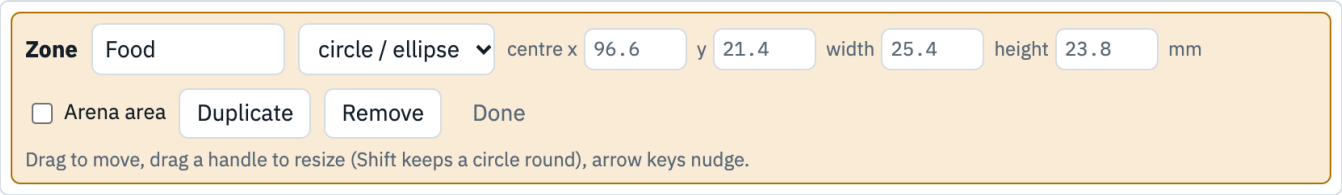
Choose **Live camera**, pick the camera and size, and press **Start camera**. Allow the camera if the browser asks. If the message under the camera says the frame rate is lower than asked for, choose a smaller size.

3 Draw the arena and set the scale

Choose round or rectangular, press **Draw arena**, and drag a box around the arena *floor*. On the **Setup** tab, type the arena's real width in mm (and height, if it isn't round or square), and set **Bees in arena**.

4 Add zones and an edge band (optional)

Press **Add zone** and drag a box. Click a zone's outline to select it: drag it, pull its handles to resize (hold `Shift` to keep a circle round), nudge it with the arrow keys, or type its name, centre and size in mm in the bar under the video. **Duplicate** makes an identical zone. An **edge band** in mm adds a zone along the wall.



A selected zone. Changing zones later is fine: the results update without another trial.

5 Set up detection

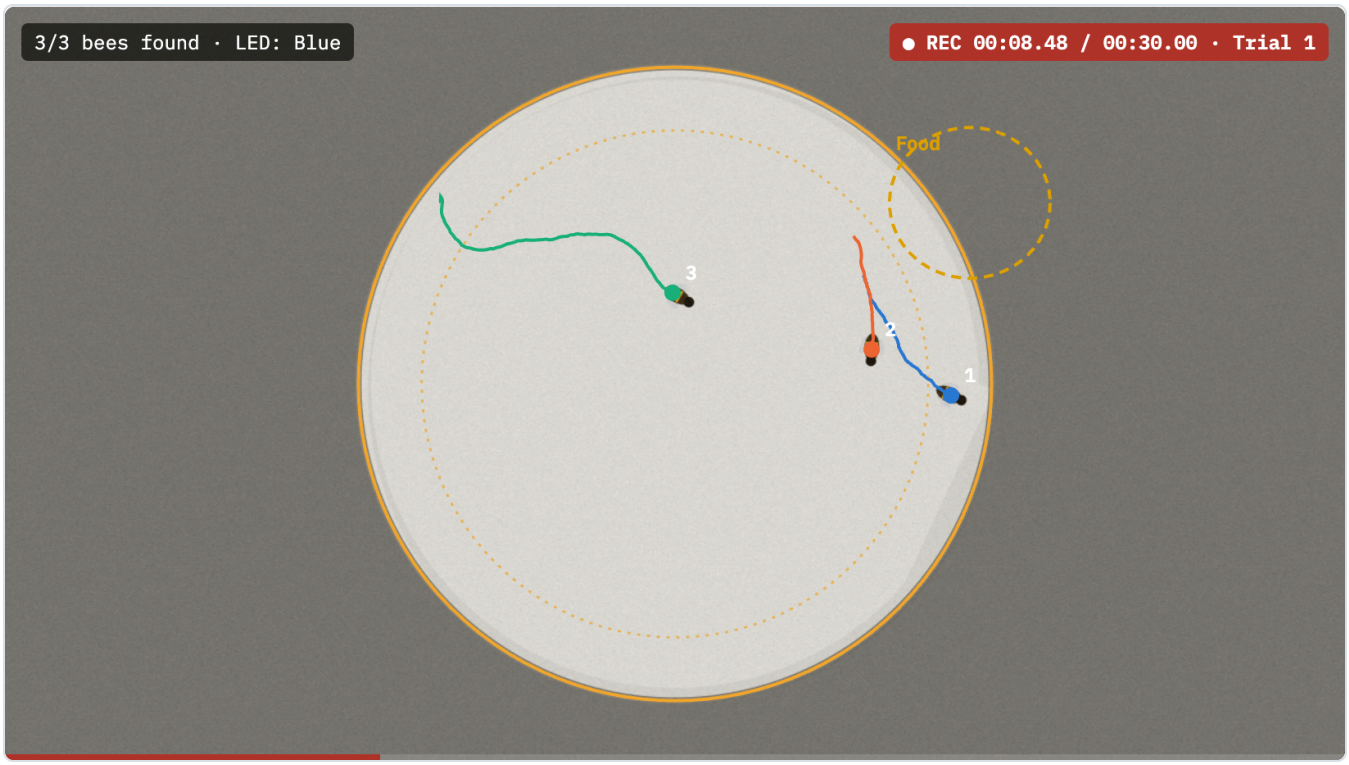
Put the bees in. On the **Detect** tab, press **Set up detection**. It learns the background while the bees move (10 s), then works out the rest. Wait for *Found all 3 bees on this frame. Ready to track.* The detected bees show in magenta. If it says something else, see Part 6.

6 Set up the LEDs (optional)

See Part 7. Leave **LEDs** unticked in the trial bar to run without them.

7 Run the trial

In the trial bar, set the **Length** in seconds (leave it empty to run until you press Stop), tick **Save video** to keep a copy of the trial, and press **Start trial**. The video shows each bee's number and track, a red timer and a progress line. **Keep the tab in front**: browsers slow down hidden tabs.



A trial running: 3 of 3 bees found, the LED condition, and the red timer.

8 Read the results, then the next trial

The results fill in when the trial ends (Part 8). Press **Start trial** again for the next one: each trial becomes its own recording, with the same arena, zones and settings. Give recordings a **Group** on the Setup tab to compare groups.

PART 5

Tracking a saved video

1. Choose **Saved video** and open a file (or paste a link to a video file, or drop it on the page).
2. Draw the arena, set its width and the number of bees, and add zones, as for a live trial.
3. Pause on a frame where every bee is visible and apart. On the Detect tab, press **Set up detection**. The background comes from frames across the whole analysis range, so bees that keep moving disappear from it.
4. To track only part of the video, go to the start and press **Start here**, then to the end and press **End here**. The range shows on the seek bar; **Whole video** resets it.
5. Press **Track bees**. It plays the video fast and samples it; for no skipped samples at all, choose *Read frames by: seeking* under Detect → Advanced (slower).

Use **New recording** for each video. It copies the arena, zones and settings, which suits a fixed camera.

Several cameras and linked arenas

Set **Cameras** to 2–4, choose a layout and a camera for each tile, and press **Start camera**: the cameras are joined into one image that works like one camera. Set each tile's **zoom** so 1 mm is the same size in every tile. For a bee

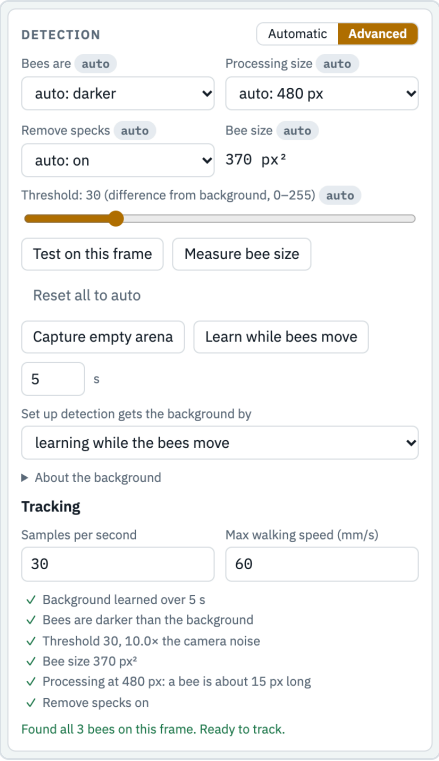
that moves between arenas, draw the arena around the whole image, draw each real arena as a zone, select it and tick **Arena area**. The results then show **Crossings**, **Transit s** (time between arenas) and the first crossing; distances between bees count only bees in the same arena.

PART 6

How detection works, and fixing it

Bee Tracker compares each frame with a picture of the arena without bees (the **background**). Pixels that differ by more than the **threshold** are bee; joined pixels form a **blob**; blobs about the size of one bee are bees, and bigger blobs are bees touching. **Set up detection** chooses these for you:

IT CHOOSES	HOW
Background	Live: the per-pixel median of frames over 10 s while the bees move. Saved video: the median of frames across the analysis range.
Darker or lighter	Whether the pixels that changed got darker or lighter than the background.
Threshold	Several times the camera's noise, then the middle of the range that finds exactly your number of bees.
Bee size	The median size of the blobs it found. Needed to split bees that touch.
Processing size	The smallest image size at which a bee is still at least 12 pixels long: faster, without losing accuracy.



Advanced

Every value it chose is here, tagged `auto`. Change one and it is tagged `set by you`: *Set up detection* keeps it from then on. **Reset all to auto** undoes that.

Capture empty arena gives the cleanest background: use it before the bees go in when you can, and choose it for *Set up detection gets the background by*.

Advanced also has the tracking settings: samples per second, the maximum walking speed (used to keep identities when bees are close), and, for saved video, how frames are read and the playback speed.

Show detection, under the video, shows the magenta mask live. Use it to check detection at any time.

When it doesn't find every bee

IT SAYS	WHAT TO DO
Found more blobs than bees	Something else is moving or casting a shadow, or the background is off. Turn on *Show detection* to see what. Fix the light, or raise the threshold under Advanced.
Found fewer bees than you have	Bees touching count as one; try again when they're apart. A bee that sat still while the background was learned became part of it: use *Capture empty arena*.
No bees found	Are the bees in view and inside the arena outline? With *learn while bees move*, they must move during those 10 s.

PART 7

LED protocols

With the live camera, each trial can play a light protocol on the two outputs (ao0, ao1) of an NI USB-6003, through the **LED helper** on the same computer. The desktop app runs the helper and connects the page for you; the LEDs tab shows *connected* and the DAQ's name. Without NI's NI-DAQmx driver, the LEDs run in simulated mode: everything works, but no LEDs light, and the panel says why.

A protocol is a list of **segments**. Each has a length, a label, and what each LED does: off, a constant level, a ramp, an exponential, a sine, a pulse train, a single pulse, random levels, or a formula of time. The protocol can repeat, and the order can be shuffled within each repeat. The LEDs tab lists the segments; **Edit protocol** widens the panel to change them.

- **Test LEDs (5 V)** turns each output on at 5 V for 1 s, one after the other, so you can check the wiring before a trial.
- Tick **LEDs** in the trial bar to play the protocol with the trial. It starts on the trial's first frame and plays on the DAQ's own clock. *End the trial when the protocol ends* stops the trial with it.
- Results are also given per LED condition (the segment labels), and the charts shade each condition.
- A **light check** box drawn over an LED the camera can see measures how late the camera sees each change.

Wiring the LEDs

Each output (**AO 0**, **AO 1**) gives a voltage against the shared **AO GND**, so every circuit returns to AO GND. The outputs give only about 5 mA, so each LED needs a driver whose current follows the voltage. The cheapest is an **op-amp current sink**: an LM358, a logic-level MOSFET (IRLZ44N) and a 1 Ω resistor, on a breadboard. LED current = control voltage / 1 Ω, so brightness follows the protocol exactly, without flicker.

CONNECT	TO
5 V supply + / −	The breadboard's red (+) and blue (−) rails
DAQ AO GND	Blue rail (the shared ground)
DAQ AO 0	15 kΩ to the op-amp's + input; 1 kΩ from that input to the blue rail (scales 5 V to 0.31 V, and keeps the LED off if the DAQ is unplugged)
Op-amp output	100 Ω to the MOSFET's gate

MOSFET source	The op-amp's − input, and 1 Ω to the blue rail
MOSFET drain / LED	Drain to the LED's −; the LED's + to the red rail
LM358 power	Red and blue rails, with 0.1 µF across them

The second LED is the same circuit on AO 1, using the chip's other op-amp. With these values, 5 V gives 312 mA. A Mean Well LDD-H is **on/off only** (it switches above about 2.5 V and dims by PWM, which bees can see as flicker): use it only for fixed-brightness on/off protocols.

> **Check your LED driver.** Levels can go up to 10 V, the DAQ's limit. If your LEDs' driver is rated for less (for example 0–5 V), keep every level in the protocol below its rating.

PART 8

Reading the results

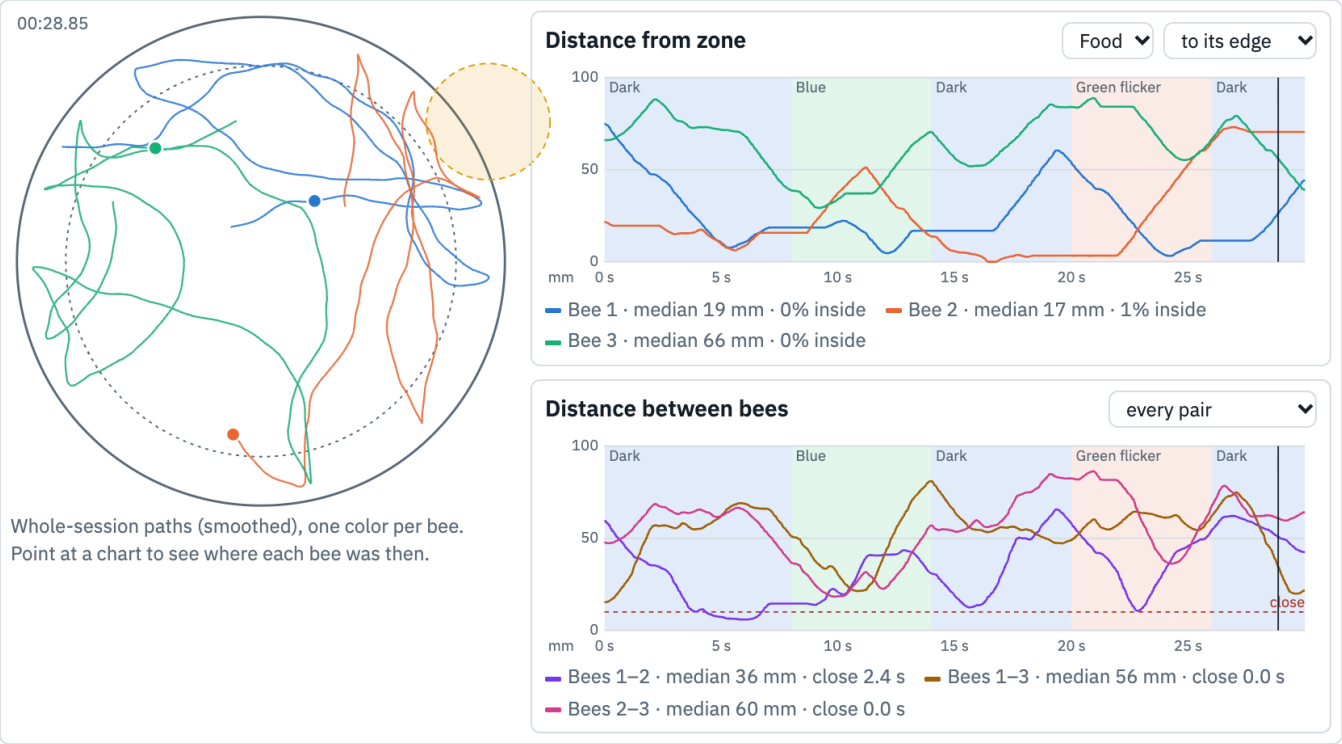
BEE	FOUND	TRACKED S	DIST MM	SPEED MM/S	MOVING	95TH MM/S	FOOD	EDGE	NEAREST MM	NEAR BEE	EN
Bee 1	100%	30.0	351	11.7	71%	26.4	0% (0)	26% (7)	30	8%	
Bee 2	100%	30.0	311	10.4	63%	26.4	1% (1)	22% (4)	33	8%	
Bee 3	100%	30.0	422	14.1	86%	26.5	0% (0)	32% (5)	47	0%	
All bees	**100%**	**90.0**	**1083**	**12.0**	**73%**	–	**0% (1)**	**27% (16)**	**37**	**5%**	

Results. One row per bee and an **All bees** row. Warnings above the table flag merges, low detection, or settings changed after tracking. *Settings* holds smoothing, the moving threshold, gaps, the shortest bout, and the distance that counts as close.

▼ By pair: time within 10 mm of each other

PAIR	CLOSE S	SHARE	ENCOUNTERS	FIRST CONTACT S	MEDIAN MM
Bees 1–2	2.4	8%	1	4.3	36
Bees 1–3	0.0	0%	0	–	56
Bees 2–3	0.0	0%	0	–	60

By pair (two or more bees): how long each pair spent close, how often, and when they first met.



Track map and charts. Left: every bee's whole path. Right: each bee's distance from a zone (pick the zone, and measure to its edge or centre), and the distance between bees (every pair, or with many bees the nearest neighbour). Shaded bands are LED conditions; the dashed red line is the *close* distance. Point at a chart, and the map shows where each bee was then; with a saved video, click to jump the video there.

Compare recordings, the second results tab, plots any measure for every recording (or every bee) by group, with Mann–Whitney or Kruskal–Wallis p-values. They are exploratory: bees in one arena are not independent.

PART 9

Saving and exporting your data

DATA TAB	WHAT YOU GET
Tracks CSV	Every sample of every bee: time, position in px and mm, smoothed position, speed, in each zone, distance to each zone and to every other bee, the LED condition.
Summary CSV	One row per bee and an *all* row per recording, for every recording, with the settings used.
Export to PER Scorer	Moving, each zone and *Near another bee* as behaviors, to use PER Scorer's time budgets and plots.
LED CSVs	Measures per LED condition, and a log of every LED segment played.
Save project (.json)	Everything: settings, tracks and results. *Open project…* brings it back.

Save the project file at the end of every session. Work is kept in the browser, but browser storage can be cleared, and the desktop app and the HTML file each keep their own.

PART 10

Good practice and troubleshooting

- Run **Set up detection** at the start of each session, and again if the light or camera moves.
- Watch the first trial with **Show detection** on, and check **Found** is near 100% for every bee.
- Keep the tab in front and the computer awake during trials.
- Keep camera size, samples per second and smoothing the same for everything you compare.
- Save the project file, and keep the trial videos, so you can re-check or re-track a trial later.

PROBLEM	WHAT TO DO
Tracking did not keep up with the camera	Lower the processing size or the samples per second under Detect → Advanced, or use a smaller camera size.
Bee found in less than 80% of samples	Detection is missing the bee: check with Show detection, and see Part 6.
"Settings changed after this trial"	The arena, bee count or detection changed since tracking. A saved video can be tracked again; a live trial keeps the settings it used.
Distances in px, not mm	Enter the arena's width in mm on the Setup tab.
LEDs say simulated	The panel says why: no NI-DAQmx driver, no DAQ plugged in, or another error. Plug in the DAQ, then restart the desktop app.
My recordings disappeared	Browser storage was cleared, or you opened the other copy (app or HTML file). Use *Open project…* with your saved file.

PART 11

Check your understanding

1. You forget to enter the arena's width. What changes in the results?
2. Set up detection says *Found 2 of 3 bees*. Give two possible reasons.
3. Three unmarked bees share an arena, and the results show 6 merges. Which row should you report, and why?
4. You move the food zone after the trial. Do you need to run the trial again?
5. Bee 2's Near bee is 25%, and the trial lasted 60 s with all of it tracked. How long was Bee 2 within the close distance of another bee?
6. Why must samples per second and smoothing be the same across groups you compare?

Answers. 1. Distances and speeds are in pixels, not mm, and the edge band can't be used. 2. Two bees were touching and counted as one, or a bee sat still while the background was learned and became part of it. 3. The All bees row: after merges, identities may have swapped, so per-bee rows are unreliable; the pooled row doesn't depend on identity. 4. No: zones are applied to the saved tracks, so the results update straight away. 5. 15 s (25% of 60 s). 6. Both change the distance measured: fewer samples or more smoothing cut corners and shorten the path.

PART 12

Quick reference card

Print this page and keep it by the rig.

Every session

1. Start the app (or open the page); fill in initials, date, experiment
2. Start camera
3. Draw arena; width in mm; bees in arena
4. Zones and edge band
5. Bees in: Detect → Set up detection
6. LEDs: connect, Test LEDs (5 V)

Every trial

1. Check the trial bar: ✓ Camera ✓ Arena ✓ Detection
2. Length, Save video, LEDs
3. Start trial; keep the tab in front
4. Check Found and the warnings

End of session: Data → Save project (.json).

Keys

`Space`	Play / pause a saved video
`←` `→`	Seek 1 s (`Shift`: 5 s)
`,` `.`	One frame back / forward
Click an outline	Select the arena, a zone or the light check
`←` `↑` `→` `↓`	Move the selected shape 1 px (`Shift`: 10 px)
`Delete`	Remove the selected shape
`Esc`	Stop drawing, or deselect

Defaults

Moving	faster than 3 mm/s
Close to another bee	within 10 mm
Smoothing	0.2 s
Test LEDs	5 V, 1 s each
LED limit	0–10 V